DEPARTMENT OF HOMELAND SECURITY
U.S. Immigration and Customs Enforcement

OMB APPROVAL NO. 1653-0054
EXPIRATION DATE: 5/31/2025

TRAINING PLAN FOR STEM OPT STUDENTS

Science, Technology, Engineering & Mathematics (STEM) Optional Practical Training (OPT)

SECTION 1: STUDENT INFORMATION (Completed by Student)

Student Name (Surname/Primary Name, Given Name): Singh, Shubhendu Kumar		Student Email Address: singhshubhendu29@gmail.com
Name of School Recommending STEM OPT: Clemson University	Name of School Where STEM Degree Was Earned: Clemson University	SEVIS School Code of School Recommending STEM OPT (including 3-digit suffix): ATL214F01380009
Designated School Official (DSO) Name and Contact Information: Clemson DSO, is@clemson.edu, 864-656-3614	Student SEVIS ID No.: N0023698774	STEM OPT Requested Period (mm-dd-yyyy): From: 01-15-2025 To: 01-14-2027

Qualifying Major and Classification of Instructional Programs (CIP) Code: Engineering, Other 14.9999

Level/Type of Qualifying Degree: Doctorate

Date Awarded (mm-dd-yyyy): 08-09-2024

Based on Prior Degree? ☐ Yes ☒ No

Employment Authorization Number: 141-357-719

SECTION 2: STUDENT CERTIFICATION

I declare and affirm under penalty of perjury that the statements and information made herein are true and correct to the best of my knowledge, information and belief. I understand that the law provides severe penalties for knowingly and willfully falsifying or concealing a material fact, or using any false document in the submission of this form.

I certify that:

1. I have reviewed, understand, and will adhere to this Training Plan for STEM OPT Students ("Plan");
2. I will notify the DSO at the earliest available opportunity if I believe that my employer is not providing me with appropriate training as delineated on this Plan;
3. I understand that the Department of Homeland Security (DHS) may deny, revoke, or terminate the STEM OPT of students whom DHS determines are not engaging in OPT in compliance with the law, including the STEM OPT of students who are not, or whose employers are not, complying with this Plan;
4. My practical training opportunity is directly related to the STEM degree that qualifies me for the STEM OPT extension; and
5. I will notify the DSO at the earliest available opportunity regarding any material changes to or deviations from this Plan, including but not limited to, any change of Employer Identification Number resulting from a corporate restructuring, any nontrivial reduction in compensation from the amount previously submitted on the Plan that is not tied to a reductio n in hours worked, any significant decrease in hours per week that I engage in a STEM training opportunity, and any decrease in hours below the 20-hours-per-week minimum required under this rule.

Signature of Student: Digitally signed by Shubhendu Kumar Singh
Date: 2024.10.29 11:54:36 -06'00'

Printed Name of Student: SHUBHENDU KUMAR SINGH Date (mm-dd-yyyy): 10-29-2024

SECTION 3: EMPLOYER INFORMATION (Completed by Employer)

Employer Name:	Street Address:	Suite:
Triad National Security, LLC	Bikini Atoll Road, Bldg SM-30	

Employer Website URL:	City:	State:	ZIP Code:
https://www.lanl.gov/	Los Alamos	NM	87545

Employer ID Number (EIN):	Number of Full-Time Employees in U.S.:	North American Industry Classification System (NAICS) Code:
82-3291283	15,000	54171

OPT Hours Per Week (must be at least 20 hours/week): 40

Start Date of Employment (mm-dd-yyyy): 01-15-2025

Compensation:

A. Salary Amount and Frequency: $94,500 Annually

B. Other Compensation (Type and Estimated Amount or Value):

1.
2.
3.
4.

SECTION 4: EMPLOYER CERTIFICATION

I declare and affirm under penalty of perjury that the statements and information made herein are true and correct to the best of my knowledge, information and belief. I understand that the law provides severe penalties for knowingly and willfully falsifying or concealing a material fact, or using any false document in the submission of this form.

I certify on behalf of the employer that this Training Plan for STEM OPT Students ("Plan") is approved and that:

1. I have reviewed and understand this Plan, and I will ensure that the supervising Official follows this Plan;
2. I will notify the DSO at the earliest available opportunity regarding any material changes to this Plan, including but not limited to, any change of Employer Identification Number resulting from a corporate restructuring, any reduction in compensation from the amount previously submitted on the Plan that is not tied to a reduction in hours worked, any significant decrease in hours per week that a student engages in a STEM training opportunity, and any decrease in hours below the 20-hours-per-week minimum required under this rule;
3. Within five business days of the termination or departure of the student during the authorized period of OPT, I will report such termination or departure to the DSO (*Note*: business days do not include federal holidays or weekend days; and an employer shall consider a student to have departed when the employer knows the student has left the practical training opportunity, or when the student has not reported for practical training for a period of five consecutive business days without the consent of the employer); and
4. I will adhere to all applicable regulatory provisions that govern this program *(see 8 CFR Part 214)*, which include, but are not limited to, the following:
   a. The student's practical training opportunity is directly related to the STEM degree that qualifies the student for the STEM OPT extension, and the position offered to the student achieves the objectives of his or her participation in this training program;
   b. The student will receive on-site supervision and training, consistent with this Plan, by experienced and knowledgeable staff;
   c. The employer has sufficient resources and personnel to provide the specified training program set forth in this Plan, and the employer is prepared to implement that program, including at the location(s) identified in this Plan;
   d. The student on a STEM OPT extension will not replace a full- or part-time, temporary or permanent U.S. worker. The terms and conditions of the STEM practical training opportunity—including duties, hours, and compensation—are commensurate with the terms and conditions applicable to the employer's similarly situated U.S. workers or, if the employer does not employ and has not recently employed more than two similarly situated U.S. workers in the area of employment, the terms and conditions of other similarly situated U.S. workers in the area of employment; and
   e. The training conducted pursuant to this Plan complies with all applicable Federal and State requirements relating to employment.

Note: DHS may, at its discretion, conduct a site visit of the employer to ensure that program requirements are being met, including that the employer possesses and maintains the ability and resources to provide structured and guided work-based learning experiences consistent with this Plan.

Signature of Employer Official with Signatory Authority: JEFFREY HEIKOOP (Affiliate) Digitally signed by JEFFREY HEIKOOP (Affiliate) Date: 2024.10.29 14:27:45 -06'00'

Printed Name and Title of Employer Official with Signatory Authority: JEFFREY MARTIN HEIKOOP, R & D MANAGER 4

Date (mm-dd-yyyy): 10/29/2024 Printed Name of Employing Organization: TRIAD NATIONAL SECURITY, LLC

SECTION 5: TRAINING PLAN FOR STEM OPT STUDENTS (Completed by Student and Employer)

Student Name *(Surname/Primary Name, Given Name):*
Singh, Shubhendu Kumar

Employer Name:
Triad National Security, LLC

EMPLOYER SITE INFORMATION

Site Name:	Site Address (Street, City, State, ZIP):
TA 51, Building 80, Room 129	Bikini Atoll Road, Bldg SM-30, Los Alamos, NM 87545
Name of Official: Jeffrey Martin Heikoop	Official's Title: R & D Manager 4
Official's Email: jheikoop@lanl.gov	Official's Phone Number: 5056953391

Note: for the remaining fields in this section, employers who already have an internal/pre-existing training plan in place may fill in the details based on that plan.

Student Role: Describe the student's role with the employer and how that role is directly related to enhancing the student's knowledge obtained through his or her qualifying STEM degree.

Student works on using various machine learning techniques, such as reinforcement learning for modeling and optimizing the behavior of complex dynamic system of a dam. During student's Ph.D. in Automotive Engineering, his research was focused on using machine learning for modeling the behavior of complex dynamic systems along with diagnosing any faults in those systems. Here as a part of his job the student gains hands on experience and gets to enhance his knowledge of machine learning and dynamics of physical systems which he studied in-depth during his Ph.D. He gets to apply the concepts learned during his Ph.D. program to real world applications.

Goals and Objectives: Describe how the assignment(s) with the employer will help the student achieve his or her specific objectives for work-based learning related to his or her STEM degree. The description must both specify the student's goals regarding specific knowledge, skills, or techniques as well as the means by which they will be achieved.

The student's goals include applying machine learning techniques to new problems important to national security, as well as adapting or developing novel techniques when required. The employer has assigned student a problem (dam operation control) focused on water security that requires optimization methods (deep reinforcement learning) student has developed in their PhD. Employer also provides data related to flood prediction as opportunity for student to build from PhD work to create new methods. To achieve goals, student will implement new software tools, run computational experiments using the data provided, and report results.

Employer Oversight: Explain how the employer provides oversight and supervision of individuals filling positions such as that being filled by the named F-1 student. If the employer has a training program or related policy in place that controls such oversight and supervision, please describe.

Employer has assigned Primary and Secondary mentors to student. Student meets twice weekly with one or both assigned mentors to ensure student maintains adequate progress. Employer holds mandatory management-level meetings with mentors and student to ensure technical and career progress is being made. Student must account for all hours worked each week through submission into employer's time tracking system.

Measures and Assessments: Explain how the employer measures and confirms whether individuals filling positions such as that being filled by the named F-1 student are acquiring new knowledge and skills. If the employer has a training program or related policy in place that controls such measures and assessments, please describe.

Employer measures and confirms that student acquires new skills through student's peer-reviewed, scientific publications, development/submission of scientific proposals, presentations to groups unfamiliar to student, and regular appraisal of the projects in which the student is involved. No formal testing is conducted to assess acquisition of new knowledge and skills, but the Employer Oversight mechanisms ensure employer monitors and re-assesses progress on these fronts. Additionally, employer provides funding for attending conferences and ample internal opportunities for learning new skills and networking with a multidisciplinary pool of staff at different career stages.

Additional Remarks *(optional):* Provide additional information pertinent to the Plan.

SECTION 6: EMPLOYER OFFICIAL CERTIFICATION

I declare and affirm under penalty of perjury that the statements and information made herein are true and correct to the best of my knowledge, information and belief. I understand that the law provides severe penalties for knowingly and willfully falsifying or concealing a material fact, or using any false document in the submission of this form.

Employer Official with Signatory Authority - I certify that:

1. I have reviewed, understand, and will follow this Training Plan for STEM OPT Students (Plan);
2. I will conduct the required periodic evaluations of the student;*
3. I will adhere to all applicable regulatory provisions that govern this program *(see 8 CFR Part 214.2(f)(10)(ii))*; and
4. I will notify the DSO regarding any material changes to or material deviations from this Plan at the earliest available opportunity, including if I believe the student is not receiving appropriate training as delineated in this Plan.

Signature of Employer Official with Signatory Authority: JEFFREY HEIKOOP (Affiliate) Digitally signed by JEFFREY HEIKOOP (Affiliate) Date: 2024.10.29 14:31:05 -06'00'

Printed Name and Title of Employer Official with Signatory Authority: JEFFREY MARTIN HEIKOOP, R & D MANAGER 4

Date (mm-dd-yyyy): 10/29/2024

PRIVACY ACT STATEMENT

AUTHORITIES: Section 101(a)(15)(F) of the Immigration and Nationality Act of 1952, as amended (INA), 8 U.S.C. 1101(a)(15)(F), Section 641 of the Illegal Immigration Reform and Immigrant Responsibility Act of 1996 (IIRIRA), Pub. L. 104-208, Div. C, 110 Stat. 3009-546 (codified at 8 U.S.C. 1372), Section 502 of the Enhanced Border Security and Visa Entry Reform Act of 2002, Pub. L. 107-173, 116 Stat. 543 (codified at 8 U.S.C. 1762) and Homeland Security Presidential Directive No. 2 (HSPD-2), authorize U.S. Immigration and Customs Enforcement (ICE) to collect the information requested in this form.

PURPOSE: The information collection on this form is used to assist in the administration of the STEM Optional Practical Training (OPT) extension so that Designated School Officials (DSO) can properly recommend the Student for and review and help coordinate his or her STEM optional practical training opportunity.

ROUTINE USES: The information collected on this form may be shared with: the individuals who signed the Plan, relevant DSOs acting as liaisons with the DHS, Federal, State, local, or foreign government entities for law enforcement purposes, Members of Congress in response to requests on the Student's behalf, or as otherwise authorized pursuant to its published Privacy Act system of records notice - Privacy Act of 1974: U.S. Immigration and Customs Enforcement, DHS/ICE-001 Student and Exchange Visitor Information System (SEVIS) System of Records (https://www.dhs.gov/system-records-notices-sorns).

DISCLOSURE: The information you provide is voluntary. However, failure to provide the information requested on this form may delay or prevent participation in a STEM OPT opportunity.

PAPERWORK REDUCTION ACT

The public reporting burden for this collection of information is estimated to average 7.5 hours per response, including time required for searching existing data sources, gathering the necessary documentation, providing the information and/or documents required, and reviewing the final collection. You do not have to supply this information unless this collection displays a currently valid Office of Management and Budget (OMB) control number. If you have comments on the accuracy of this burden estimate and/or recommendations for reducing it, send them to: U.S.Immigration and Customs Enforcement, Office of Policy, 500 12th Street SW, Washington, D.C. 20536

*See evaluation forms that follow for student's first evaluation, to occur before the one year anniversary of the start date of the student's STEM OPT employment authorization, and final program evaluation.

EVALUATION ON STUDENT PROGRESS

Provide a self-evaluation of your performance, using the measures previously identified, in applying and acquiring new knowledge, skills, and competencies identified in the Training Plan for STEM OPT Students. Discuss accomplishments, successful projects, overall contributions, etc., during this review period. Address whether there are any modifications to the objectives and goals for projects, or new areas for skill and competency development.

Range of Evaluation Dates: From (mm-dd-yyyy): _____ To (mm-dd-yyyy): _____

Signature of Student: _____

Printed Name of Student: _____ Date (mm-dd-yyyy): _____

Signature of Employer Official with Signatory Authority: _____

Printed Name of Employer Official with Signatory Authority: _____ Date (mm-dd-yyyy): _____

FINAL EVALUATION ON STUDENT PROGRESS

Provide a self-evaluation of your performance, using the measures previously identified, in applying and acquiring new knowledge, skills, and competencies identified in the Training Plan for STEM OPT Students. Discuss accomplishments, successful projects, overall contributions, etc., during this review period. Address whether there are any modifications to the objectives and goals for projects, or new areas for skill and competency development.

Range of Evaluation Dates: From (mm-dd-yyyy): _____ To (mm-dd-yyyy): _____

Signature of Student: _____

Printed Name of Student: _____ Date (mm-dd-yyyy): _____

Signature of Employer Official with Signatory Authority: _____

Printed Name of Employer Official with Signatory Authority: _____ Date (mm-dd-yyyy): _____